inputs cause an LLM to output harmful text. To mitigate this problem, developers should use tools to generate inputs which result in undesired behavior and train against these adversarial examples (Zhang and Li, 2019; Ziegler et al., 2022; Perez et al., 2022a; Casper et al., 2023b). Anomaly detection techniques (Omar et al., 2013) can also be useful for flagging abnormal inputs likely to trigger bad behavior. Ensuring the security of important AI training runs against malicious human evaluators and/or outside cybersecurity threats will also be valuable.

Risk assessment and auditing. Although training processes should be crafted to produce models that are safe by design, evaluations are another layer of defense. Passing an evaluation is not proof of safety, but as is the case in almost every safety-critical industry, rigorous evaluations of capabilities and risks helps to spot hazards and establish trust. In practice, this should involve both in-house and second-party evaluations (OpenAI, 2023; ARC, 2022; Perez et al., 2022b). As with adversarial training for robustness, the development of improved red teaming techniques will be important (Perez et al., 2022a; Casper et al., 2023b).

Interpretability and model editing. Generating human-understandable explanations for the behavior of AI systems is currently an unsolved problem. Progress in explainability and interpretability could help verify hypotheses about how models make decisions (Geiger et al., 2023), including whether the decision-making process is trustworthy. In this way, it could be possible to gain confidence that models will (or will not) behave in a safe way without necessarily conducting extensive testing of the models (Jacovi et al., 2021). Red-teaming can also be complemented by interpretability techniques (Rastogi et al., 2023; Räuker et al., 2023), especially for purposes of identifying adversarial inputs (Ziegler et al., 2022; Casper et al., 2023c;a) or anomalous inputs (Pang et al., 2021). In another direction, better understanding the internal mechanisms of models can aid in directly editing model weights or intervening on internal activations in order to improve truthfulness (Li et al., 2023b), modify a model's factual knowledge (Meng et al., 2023; 2022; Hernandez et al., 2023; Hase et al., 2023), or otherwise steer model behavior (Cui et al., 2022).

5 Governance and Transparency

Social scientists and policymakers have increasingly focused on the need for governance frameworks to develop and deploy AI systems responsibly. Across historical contexts, a hallmark of mature scientific fields is the open sharing of research findings (Shapin and Schaffer, 2011) to allow experts to understand progress (Gilbert and Loveridge, 2021). Below we overview components of an RLHF governance agenda, including outstanding questions and risk dimensions.

Incentives and requirements for safety. Competition between labs can generate harmful race dynamics (Dafoe, 2018) because of tradeoffs between competitiveness and caution. This suggests a role for governance in promoting a healthier environment for safe AI research, development, and deployment (Dafoe, 2018; Perry and Uuk, 2019; Falco et al., 2021; Cihon, 2019; Anderljung et al., 2023). Governance in this form could involve mandates for independent auditing, evaluations, and certification (Shavit, 2023; Mökander et al., 2023; ARC, 2022; Hadfield and Clark, 2023; Shevlane et al., 2023); monitoring for post-deployment problems (Hendrycks and Gimpel, 2016); influence over resources including hardware and data (Brief, 2020; Chan et al., 2023a); and prohibiting deployment unless critical standards are met, as in the case of the U.S. Food and Drug Administration's oversight of clinical trials for testing potential new treatments (Junod, 2008).

Transparency and auditing. A sustained commitment to transparency would make the existing RLHF research environment more robust from a safety standpoint. First, the disclosure of some details behind large RLHF training runs would clarify a given organization's norms for model scrutiny and safety checks. Second, increased transparency about known efforts to mitigate risks could improve safety incentives and suggest methods for external stakeholders to hold companies accountable. Third, and most relevant for the present paper, transparency would improve the AI safety community's understanding of RLHF and support the ability to track technical progress on its challenges. Some level of disclosure is a precondition to evaluate the viability of the technical RLHF safety agenda over time and allow for community contribution to it. For all of these reasons, working to incorporate transparency standards into an AI governance framework will be important (Larsson and Heintz, 2020; Anderljung et al., 2023). It is possible that public disclosure of details critical to the development of model capabilities might lead to the unwanted proliferation of AI technologies

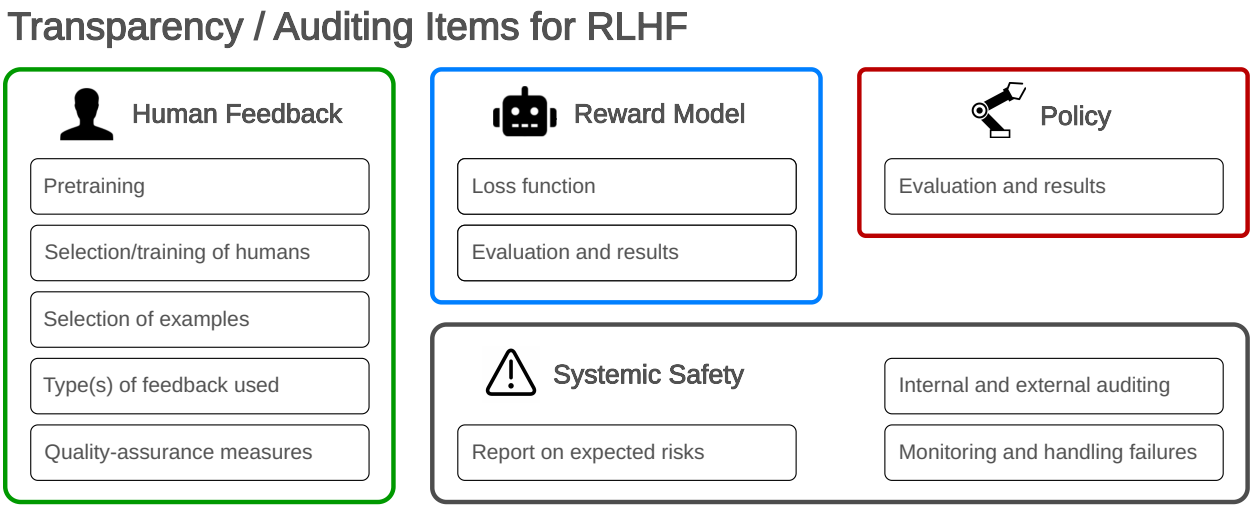

Figure 4: **Details behind an implementation of RLHF that, if disclosed, could be indicative of risks.** See Section 5 for a complete discussion. Companies using RLHF to train models for high-stakes or safety-critical applications should maintain transparency with the public and/or auditors about key details of their approach.

that could be misused. However, detailing safety measures will often not require divulging implementable details, and when it does, private disclosure to second-party auditors (Mökander et al., 2023; ARC, 2022; Hadfield and Clark, 2023; Shevlane et al., 2023) offers a solution.

As more specific policy prescriptions are beyond our scope, we encourage elaboration on these topics as part of a future research agenda. Below, however, we outline specific types of details that, if disclosed, could be indicative of risks and should be accounted for when auditing AI systems developed using RLHF. See also Figure 4.

Human feedback details:

- **A description of the pretraining process including details about what data was used** to make apparent possible biases that pretraining can cause.
- **How human evaluators were selected and trained** to provide information about risks of evaluators being malicious, unrepresentative, or incapable.
- **The process by which examples were selected to obtain feedback** to invite scrutiny about their representativeness and whether sufficient adversarial training was used. If examples were crowdsourced from a publicly-available application, details about what measures were taken to avoid data poisoning attacks should be provided.
- **The type(s) of human feedback used** (e.g., binary comparisons, scalar feedback, etc.) to suggest what risks might be caused by insufficiently abundant or rich feedback.
- **A report on measures taken for quality assurance in feedback collection and inter-rater consistency** to ensure that effective quality control measures were taken.

Reward model details:

- **The loss function used to fit the reward model and how disagreement was modeled** (e.g., as noise) to help with analyzing the degree of misspecification when fitting the reward model.
- **A report on reward model evaluation and results** to suggest possible problems from a misaligned reward model. The evaluation should involve red teaming.

Policy details:

- **A report on policy evaluation and results** to suggest possible troubles from a misaligned policy. The evaluation should involve red teaming and include assessment for risky capabilities (e.g., the ability to deceive a human).

Systemic safety measures

- **A report on internal and external audits and red teaming** to ensure accountability and disclose risks that are identified.
- **A report on expected risks and anticipated failure modes** to ensure accountability.
- **Plans for monitoring and correcting failures that emerge** to support post-deployment safety.

How these types of risks should be documented remains an area of active work in AI governance. Similar questions have been asked in an investigation by the US Federal Trade Commission into OpenAI (FTC, 2023) but in response to problems with ChatGPT rather than proactively. Salient documentation proposals focus on regular reporting of reward components (Gilbert et al., 2022) and the ability to compare the capabilities of language models according to standard benchmarks (Liang et al., 2022a). For the longer term, incorporating beneficial standards for safety and transparency into norms and regulations affecting AI is an ongoing challenge.

Concerns for social and economic equity. Although this paper has focused on technical challenges with RLHF, there are social and economic ones as well which governance and industry should work to address. For example, OpenAI has paid Kenyan knowledge workers at a rate of less than $2 USD per hour (Perrigo, 2023) for work which was mentally and emotionally demanding (Hao, 2023). Human subjects used in RLHF research should not be systematically selected simply for their availability or low cost (National Commission for the Protection of Human Subjects, 1978). Costs, benefits, and influence over RLHF models should be equitably distributed across different communities (Whittlestone et al., 2021; Eloundou et al., 2023). There is an additional possibility that powerful AI systems will be highly profitable and serve to concentrate large amounts of wealth and power into the hands of a few (O'Keefe et al., 2020; Chan et al., 2023b). Thus, policies that address inequalities and protect vulnerable populations (e.g. impacted communities, whistleblowers) will be increasingly important.

6 Discussion

While some problems with RLHF are tractable, others are fundamental. Technical progress in some respects is tractable, and this room for progress should be seen as a cause for concerted work and optimism. Even some of the fundamental problems that we overview can be alleviated with improved methodology even though they cannot be fully solved by RLHF. However, the fundamental nature of these problems requires that they be avoided or compensated for with non-RLHF approaches. Hence, we emphasize the importance of two strategies: (1) evaluating technical progress in light of the fundamental limitations of RLHF and other methods, and (2) addressing the sociotechnical challenges of aligning to human values by committing to both defense-in-depth safety measures and openly sharing research findings with the wider scientific community.

RLHF = Rehashing Lessons from Historical Failures? RLHF offers new capabilities but faces many old problems. Its use by Christiano et al. dates to 2017, and the individual components of it (preference elicitation, fitting a reward model, and policy optimization) have a history of technical and fundamental challenges in the fields of human-computer interaction and AI safety. In 2023, RLHF was described by the first author of Christiano et al. (2017) as a "basic solution" intended to make it easier to "productively work on more challenging alignment problems" (Christiano, 2023).[3] Some challenges and questions that we have

[3]Christiano (2023) mentions debate (Irving et al., 2018) and recursive reward modeling (Leike et al., 2018) as examples of 'more challenging alignment problems.' See also an outline of proposals in Hubinger (2020).